

MBA RAPID FIRE

CAT 2025 - FINAL LAP

— FOR MBA ASPIRANTS —

Lecture - 01

Arithmetic - 1

By- RAHUL BATHLA SIR



CAT 99.95 %ile

TOPICS TO BE COVERED

1

Important Concepts + Shortcuts

2

PRACTICE QUESTIONS

100 %ile



SATURDAY → YT

12:00

LIVE MOCK

11/22

EASY - MODERATE

BASIC CONCEPTs

95%ile - 100%ile

5/11

→ MODER → DIFFICULT

DIFFICULT PRACTICE PROBLEMS

CHEAT SHEETS

A horizontal glowing blue pen with gold-colored tips, positioned behind the word 'PERCENTAGE'.

PERCENTAGE

$$A\% \text{ of } B = B\% \text{ of } A = AB/100$$

$$30\% \text{ of } 240 = 72$$

$$20\% \text{ of } 80 = 16$$

$$40\% \text{ of } 70 = 28$$

$$10\% \text{ of } 242 = 24.2$$

$$1\% \text{ of } 242 = 2.42$$

$$32\% \text{ of } 420$$

$$30\% \text{ of } 420 + 2\% \text{ of } 420$$

$$126 + 8.4$$

$$\underline{134.4}$$

PERCENTAGE - 2



PERCENTAGE

FRACTION

PERCENTAGE

FRACTION

11.11%

$\frac{1}{9}$

09.09%

$\frac{1}{11}$

22.22%

$\frac{2}{9}$

18.18%

$\frac{2}{11}$

33.33%

$\frac{3}{9}$

27.27%

$\frac{3}{11}$

44.44%

$\frac{4}{9}$

36.36%

$\frac{4}{11}$

55.55%

$\frac{5}{9}$

45.45%

$\frac{5}{11}$

66.66%

$\frac{6}{9}$

54.54%

$\frac{6}{11}$

77.77%

$\frac{7}{9}$

63.63%

$\frac{7}{11}$

88.88%

$\frac{8}{9}$

72.72%

$\frac{8}{11}$

$14.28\% = \frac{1}{7}$

81.81%

$\frac{9}{11}$

90.90%

$\frac{10}{11}$

PERCENTAGE	FRACTION
12.5 %	$\frac{1}{8}$
25%	$\frac{2}{8}$
37.5%	$\frac{3}{8}$
50%	$\frac{4}{8}$
62.5%	$\frac{5}{8}$
75%	$\frac{6}{8}$
87.5%	$\frac{7}{8}$
100%	$\frac{8}{8}$ or 1

LORD



UNITARY METHOD

45% {
Car EMI = 5% of Salary
HOME EMI = 25% of Salary
LEISURE SPENDING = 15% of Salary

Saving = 55000 Rs

Salary = ?? 100000

55% → 55000

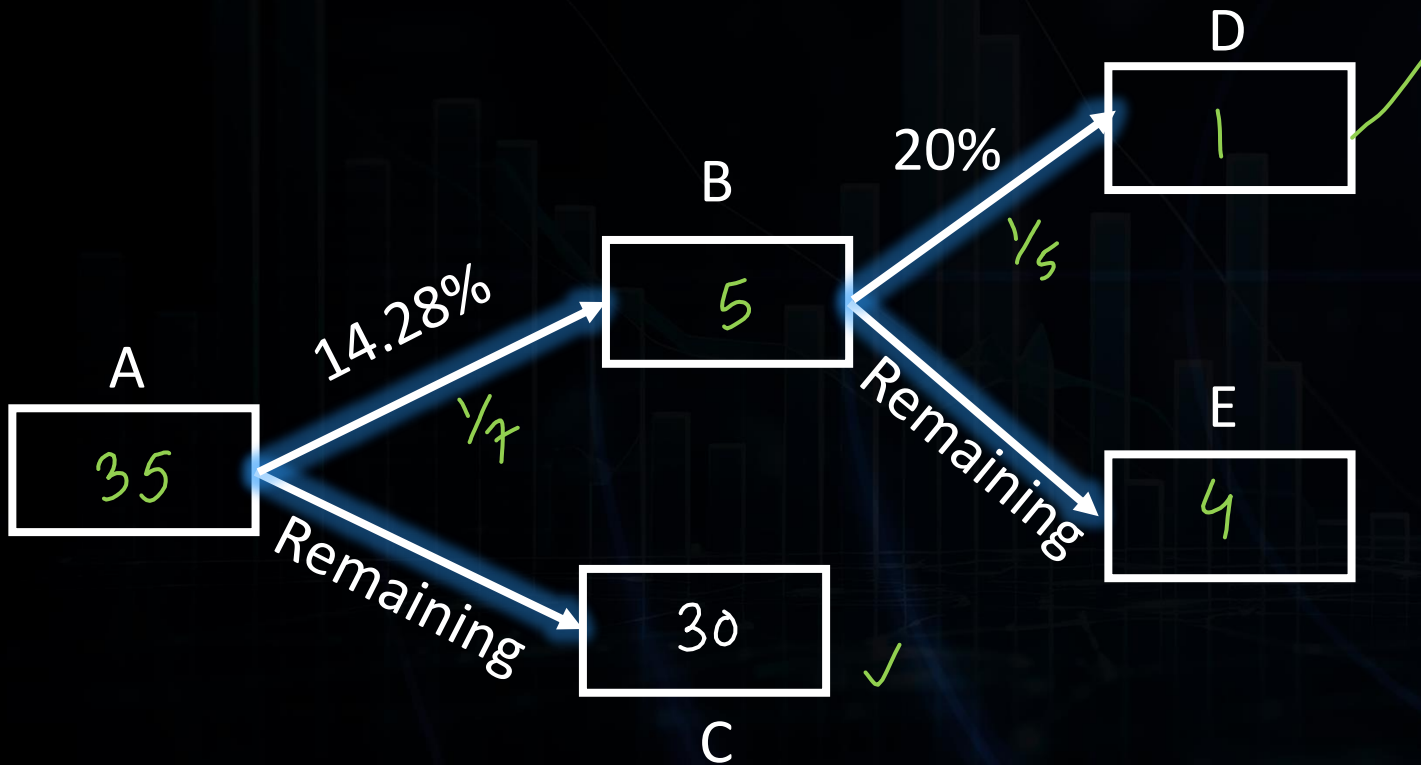
100% →

LORD

UNITARY METHOD



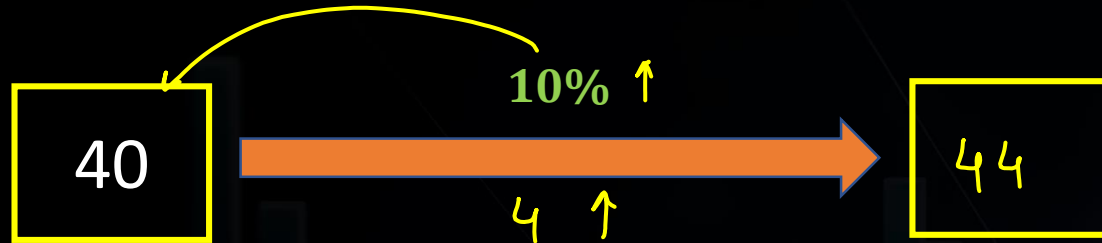
A gave 14.28% to B and remaining to C, B gave 20% to D and remaining to E.



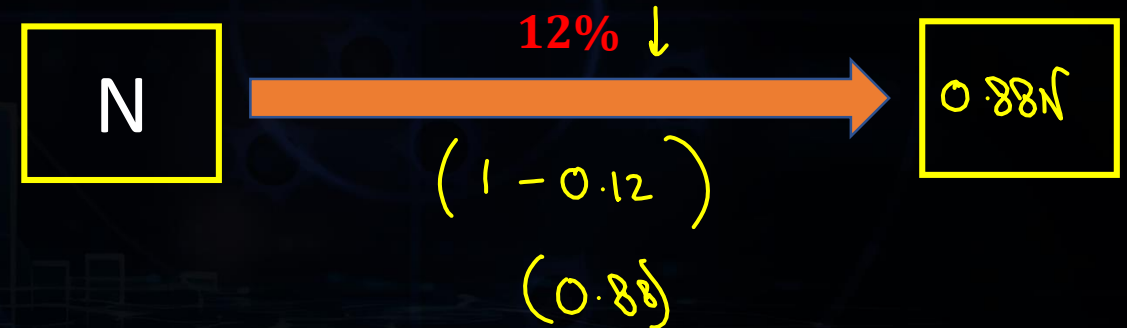
Diff b/n C & D = 58

find E = 8

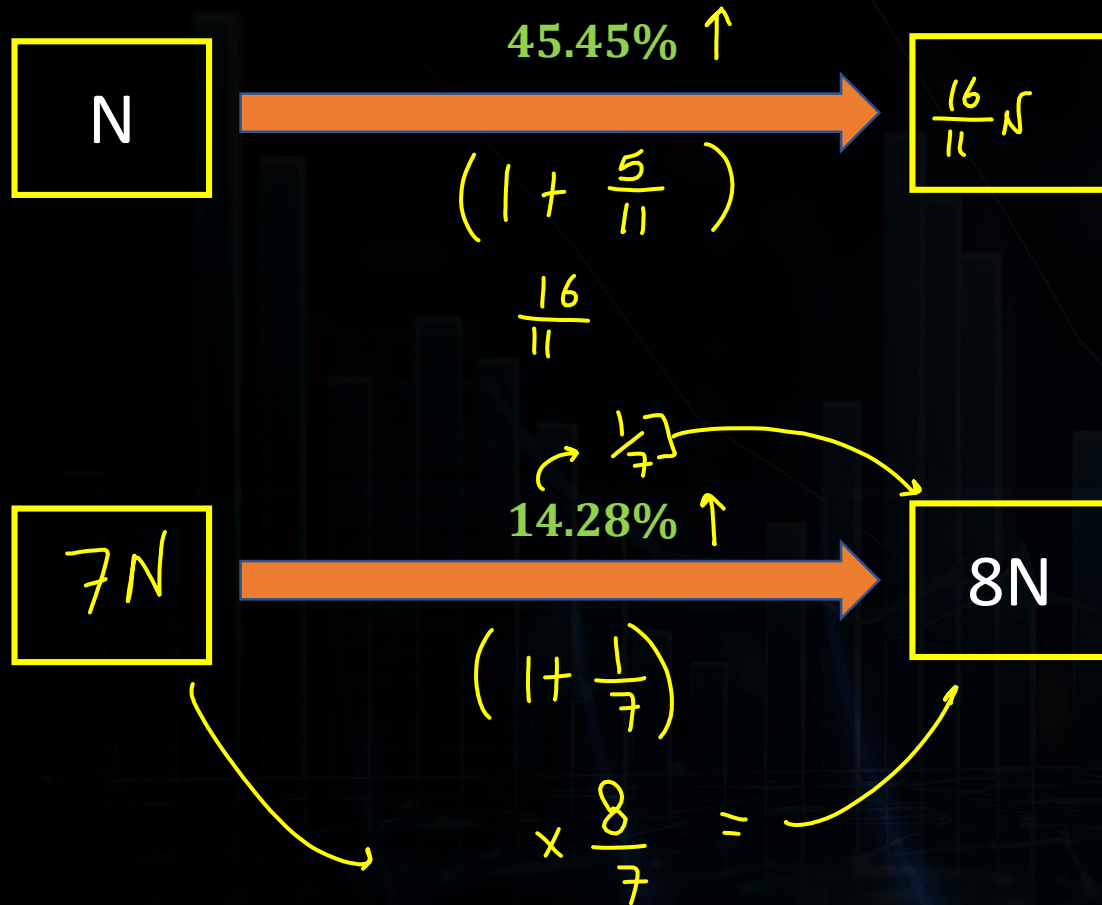
A% more than B 'or' B increased by A%



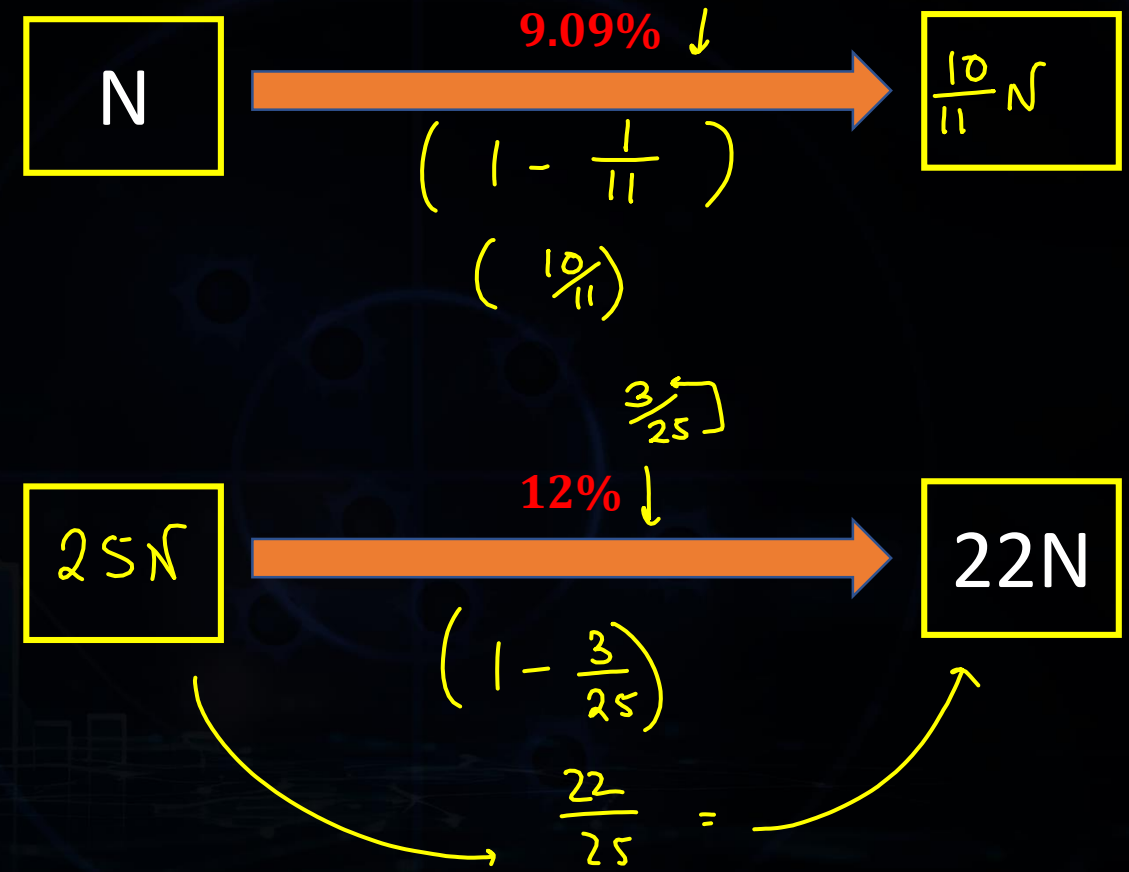
A% less than B 'or' B decreased by A%



A% more than B 'or' B increased by A%



A% less than B 'or' B decreased by A%



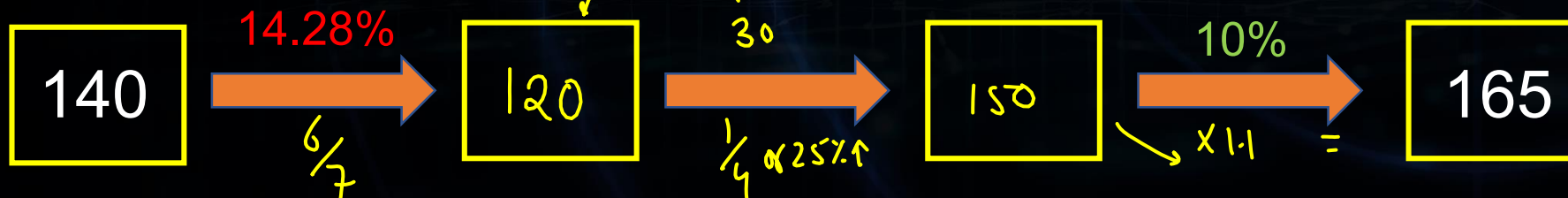
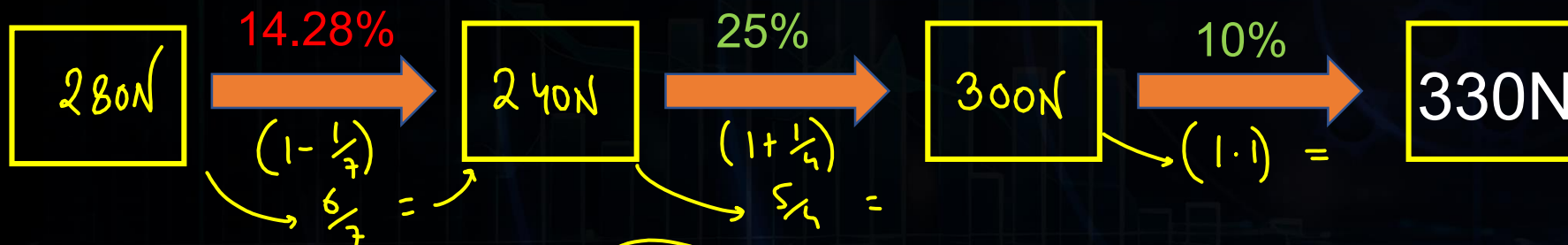
PERCENTAGE - 5



$$A + B + \frac{AB}{100} = \text{Net \%}$$



$$N[0.8][1.3][1.1] =$$



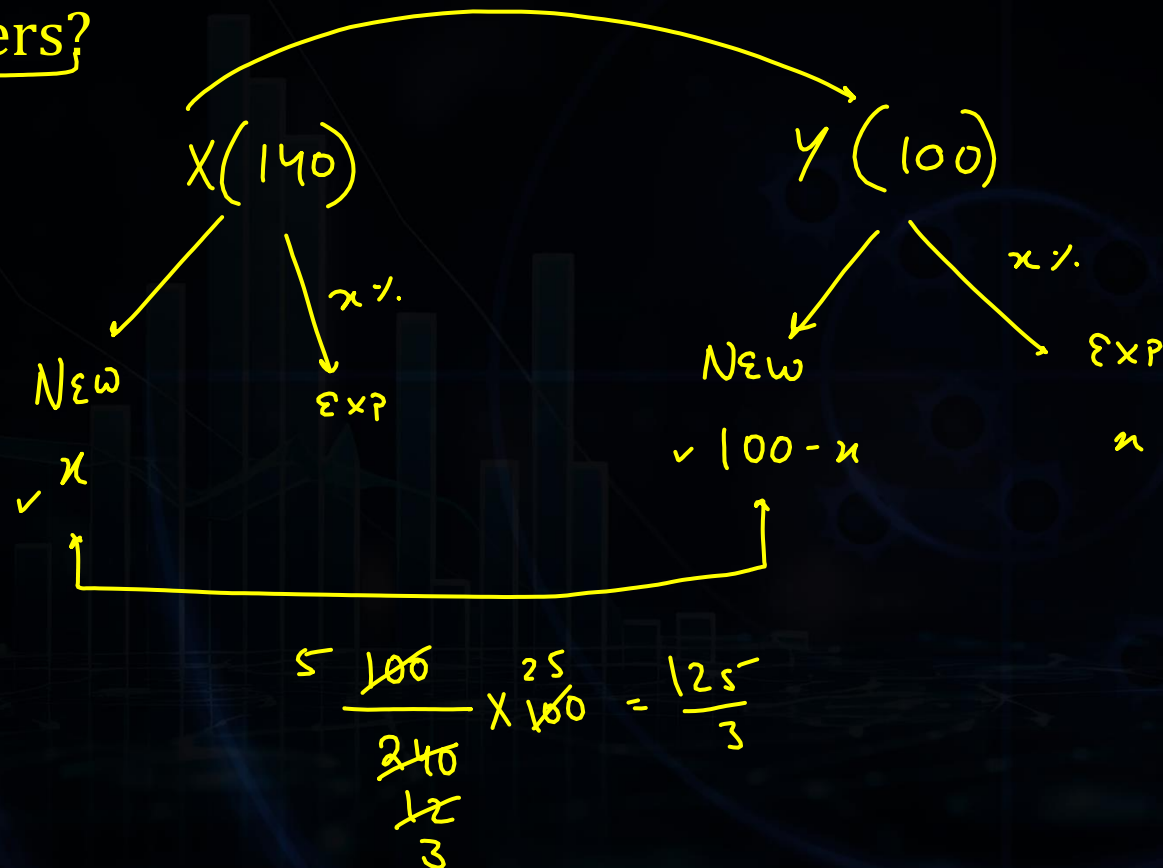
Team X has 40% more players than Team Y. The number of experienced players in Team Y is equal to the number of new players in Team X. If the percentage of experienced players is the same in both teams, what percentage of all players (in both teams combined) are new players?

A. 41%

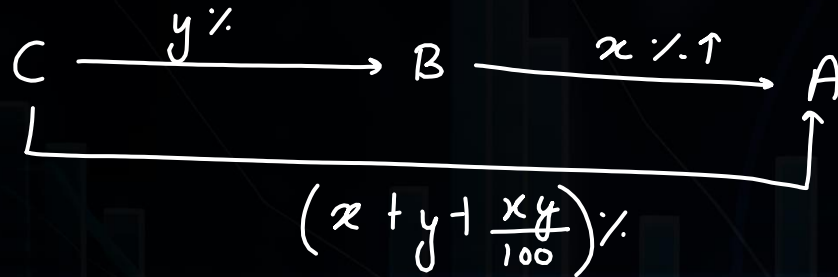
B. 41.33%

C. 41.67% ✓

D. 42%



A is $x\%$ more than B and B is $y\%$ more than C. If A is $a\%$ more than C. Find the value of $100(a_{max} - a_{min})$ where a_{max} and a_{min} represents the maximum value and minimum value of a given that $x + y = 20$ where x and y are positive integers.



$$1 - \frac{19}{100}$$

$$= \frac{81}{100}$$

$$100\left(\frac{81}{100}\right) - 81$$

$$\left(20 + \frac{xy}{100}\right)\% = a\%$$

$$20 + \frac{100}{100} \quad \bigg| \quad 20 + \frac{19}{100}$$

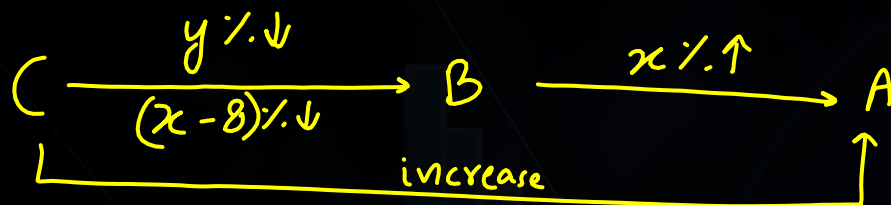
$$21$$

$$x + y = 20$$

$$\text{Max } xy = (10)(10) = 100$$

$$\text{Min } xy = (1)(19) = 19$$

A is $x\%$ more than B and B is $y\%$ less than C. ~~If C is $z\%$ more than A.~~ If $x - y = 8$, and $A > C$ and $B < C$. Find the number of possible values of x , if x is an integral number.



$$y = x - 8$$

$$-y + x + \frac{(x)(-y)}{100} > 0$$

$$x - 8 > 0$$

$$32 \geq \boxed{x > 8}$$

$$x - y - \frac{xy}{100} > 0$$

$$8 > \frac{xy}{100}$$

$$800 > xy$$

$$800 > x(x-8)$$

$$800 > x^2 - 8x$$

$$16 + 800 > x^2 - 8x + 16$$

$$816 > (x-4)^2$$

$$x - 4 = 28$$

$$\boxed{x = 32}$$

A fruit seller has a total of 187 fruits consisting of apples, mangoes and oranges. The number of apples and mangoes are in the ratio 5 : 2. After she sells 75 apples, 26 mangoes and half of the oranges, the ratio of number of unsold apples to number of unsold oranges becomes 3 : 2. The total number of unsold fruits is 15

$x = 21$
 $y = 26$

A	M	O	
$5x$	$2x$	$2y$	187
↓ -75	↓ -26	↓ -y	121
$5x - 75$	$2x - 26$	y	

$7x + 2y = 187$
66 ✓

$$\frac{5x - 75}{y} = \frac{3}{2}$$

$$10x - 150 = 3y$$

$$10x - 3y = 150$$

+3 (15	0	+10
18	10	+10
+3 (21	20	+10

Q.

QUESTION - 5



A fruit seller has a stock of mangoes, bananas and apples with at least one fruit of each type. At the beginning of a day, the number of mangoes make up 40% of his stock. That day, he sells half of the mangoes, 96 bananas and 40% of the apples. At the end of the day, he ends up selling 50% of the fruits. The smallest possible total number of fruits in the stock at the beginning of the day is

M	B	A
$2x$	y	$5z$
$\downarrow x$	$\downarrow 96$	$\downarrow 2z$
x	$y - 96$	$3z$

$$5x \quad 5(68) \quad (340) \checkmark$$

$$2.5x$$

$$40\% \rightarrow \frac{2}{5} \quad \begin{matrix} 40 \\ 40x \\ 1.2 \end{matrix} \quad \begin{matrix} 100 \\ 100x \\ 1.2 \end{matrix}$$

$$40\% \rightarrow \left(\frac{2}{5} \right)$$

$$5x \quad (2x) \quad (5x)$$

$$y + 5z = 3x$$

$$y + 3z - 96 = 1.5x$$

$$\begin{aligned} z &= 3 \\ x &= 68 \end{aligned}$$

$$(2z + 96 = 1.5x) \times 2$$

$$4z + 192 = 3x$$

$$4(2 + 17) = 3x$$

CHEAT SHEETS

RATIO



$$A : B = 1 : 4$$

$$A(x) \quad B(4x)$$

$$A : B : C = 1 : 2 : 3$$

$$A : T = 1 : 6$$

$$B : T = 2 : 6$$

$$C : T = 3 : 6$$

$$A : B : C = \frac{1}{4} : \frac{2}{3} : \frac{3}{2} \Bigg] \times 12$$

$$A : B : C = 3 : 8 : 18$$

$$A : B = 2 : 3$$

$$C : B = 7 : 5$$

$$2A = 5B$$

$$A : B : C$$

$$\begin{array}{ccc} 2 & 3 & \\ & 5 & 7 \\ \hline \end{array}$$

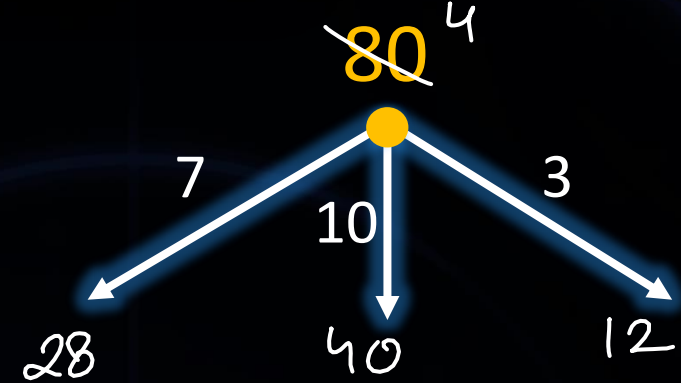
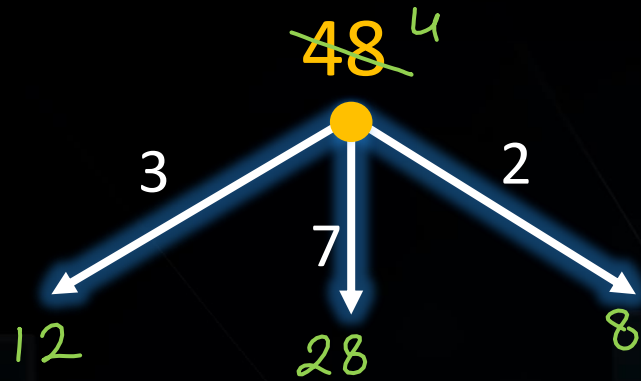
$$10 : 15 : 21$$

$$A = 5x$$

$$B = 2x$$

$$2A = 5B = 3C$$

$$A = 15x \quad B = 6x \quad C = 10x$$



$$A : B = 1 : 4$$

$$B : C = 2 : 3$$

$$C : D = 4 : 7$$

$$D : E = 7 : 3$$

$$\frac{A}{E} = \frac{1}{4} \times \frac{2}{3} \times \frac{4}{7} \times \frac{7}{3}$$

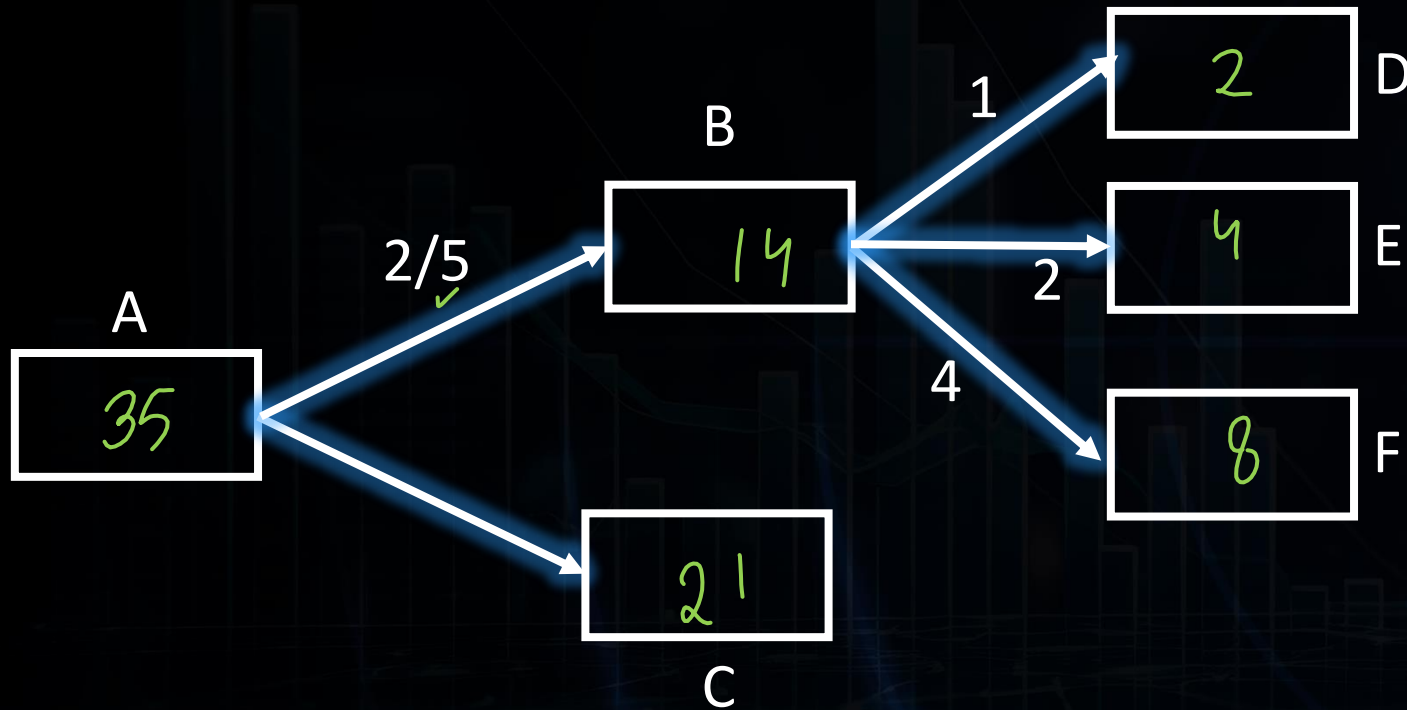
$$= \frac{2}{9}$$

LORD



UNITARY METHOD

D:E:F = 1:2:4



$$C - D = 19 \quad \text{57}$$
$$F - E = 12 \quad \text{4}$$

$$\frac{\sqrt{1+x} - \sqrt{1-x}}{\sqrt{1+x} + \sqrt{1-x}} = \frac{1}{4}$$

$$\frac{2\sqrt{1+x}}{+2\sqrt{1-x}} = \frac{5}{+3}$$

$$\frac{\sqrt{1+x}}{\sqrt{1-x}} = \frac{5}{3}$$

$$\frac{1+x}{1-x} = \frac{25}{9}$$

$$\frac{x}{x} = \frac{34}{16}$$

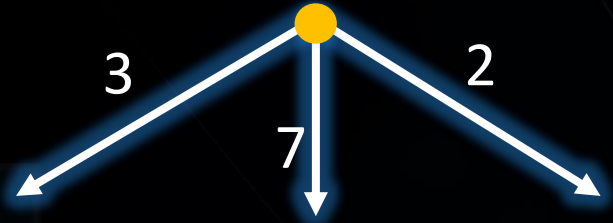
$$\frac{[\sqrt{1+x} - \cancel{\sqrt{1-x}}] + [\sqrt{1+x} + \cancel{\sqrt{1-x}}]}{[\cancel{\sqrt{1+x}} - \sqrt{1-x}] - [\cancel{\sqrt{1+x}} + \sqrt{1-x}]} = \frac{1+4}{1-4}$$

$$\left(\frac{a}{b} = \frac{c}{d} \right)$$

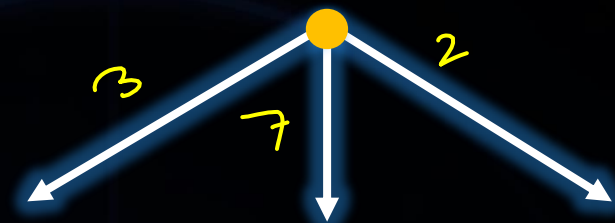
$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$$x = \frac{16}{34} = \frac{8}{17}$$

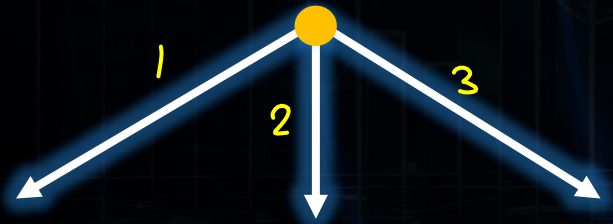
INVESTMENT



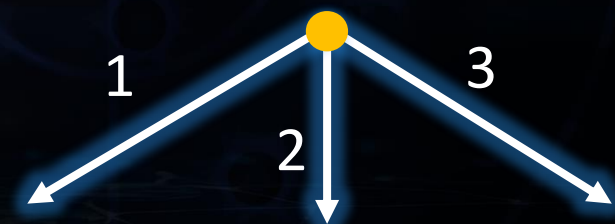
PROFIT



INVESTMENT



PROFIT



PROFIT

$$A$$

$$I_A \times T_A$$

$$B$$

$$I_B \times T_B$$

$$C$$

$$I_C \times T_C$$

PROFIT

$$A$$

$$\frac{20000}{4} \times \frac{8}{2}$$

$$B$$

$$\frac{25000}{5} \times \frac{4}{1}$$

$$C$$

$$\frac{10000}{2} \times \frac{12}{3}$$

8

:

5

:

6

$$x \propto y$$

$$x \propto \frac{1}{z^2}$$

$$x = 20 \quad \text{if } y = 10 \text{ \& } z = 2$$

MAKING FORMULA

$$x = \frac{ky}{z^2}$$

FINDING K

$$\frac{20}{2} = \frac{k(10)}{4}$$

$$8 = k$$

SOLVING UNKNOWN

$$x = \frac{8y}{z^2}$$

Q.

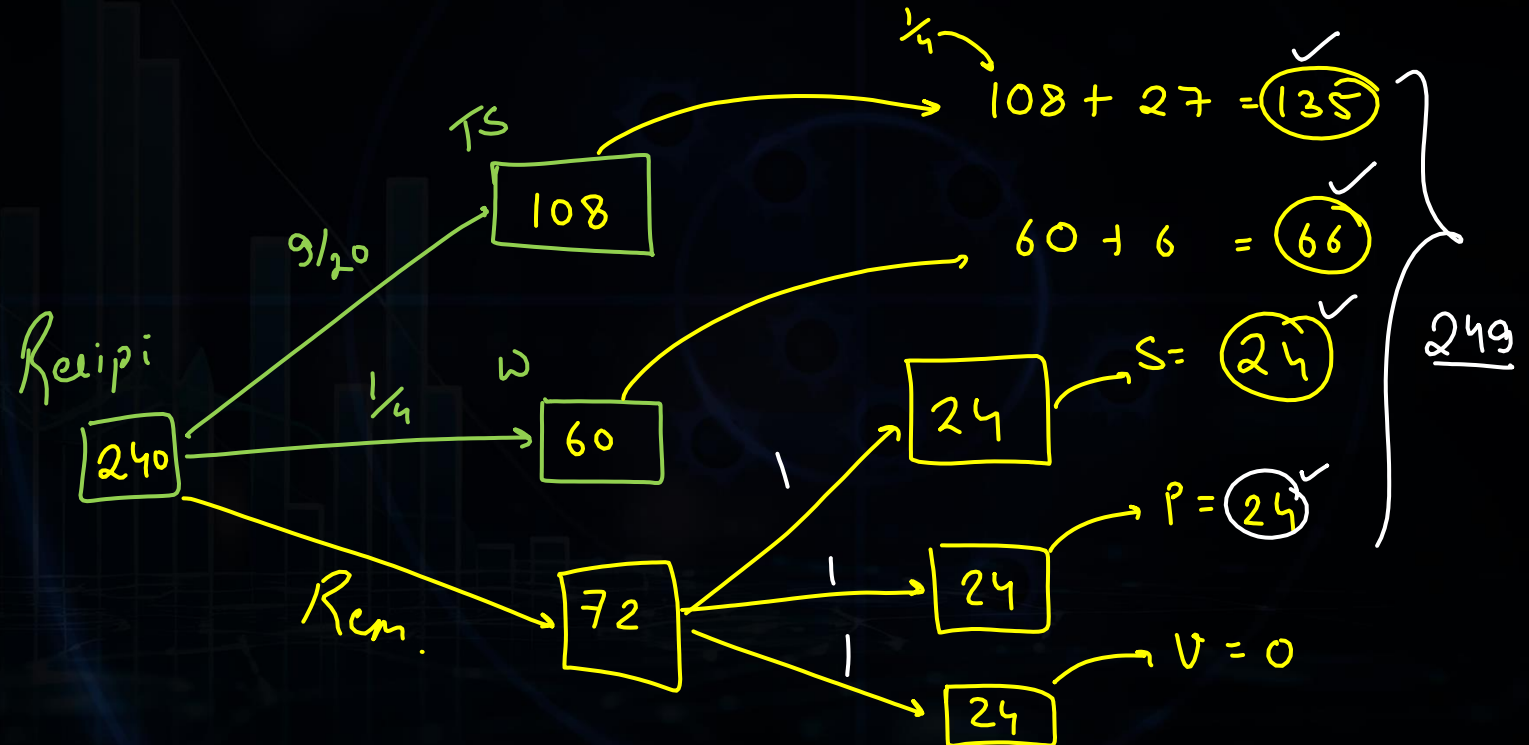
QUESTION - 6



A tomato soup recipe is such that the tomato syrup is 45% of the entire ingredient and $\frac{1}{4}$ th of the proportion is water. The remaining part is equally distributed in salt, pepper and vinegar. Amrit tried to copy the above recipe but mistakenly added 25% more syrup, 10% more water and forgot to add the vinegar. Find what is the proportion of salt in the entire mixture.

A. $\frac{7}{81}$ B. $\frac{8}{83}$ ✓✓C. $\frac{9}{80}$ D. $\frac{6}{81}$

$$\frac{24}{249} = \frac{8}{83}$$



Q.

QUESTION - 7



Mr. X has 'a' 10 Rs. Notes, 'b' 20 Rs. notes, 'c' 50 Rs notes and 'd' 100 Rs notes. If $a : b = 2 : 1$ and $c : d = 3 : 4$ and $ac - bd = 12$, Find the amount Mr. X has, given that a, b, c and d are distinct integer and the difference in the number of a unique currency notes of any two types is a one digit number,

Handwritten work for the problem:

From $a : b = 2 : 1$, we have $a = 2x$ and $b = x$.
 From $c : d = 3 : 4$, we have $c = 3y$ and $d = 4y$.

Substituting into $ac - bd = 12$:

$$(2x)(3y) - (x)(4y) = 12$$

$$6xy - 4xy = 12$$

$$2xy = 12$$

$$xy = 6$$

Since x and y are distinct integers, the possible pairs (x, y) are (1, 6), (2, 3), (3, 2), and (6, 1).

Calculating the total amount for each pair:

(x, y)	a	b	c	d	Amount (Rs)
(1, 6)	20	10	180	240	450
(2, 3)	40	20	90	120	790
(3, 2)	60	30	60	80	780
(6, 1)	120	60	30	40	750

The only amount that is a multiple of 10 and has a one-digit difference in the number of notes of any two types is 790.

Handwritten work for the problem:

From $a : b = 2 : 1$, we have $a = 2x$ and $b = x$.
 From $c : d = 3 : 4$, we have $c = 3y$ and $d = 4y$.

Substituting into $ac - bd = 12$:

$$(2x)(3y) - (x)(4y) = 12$$

$$6xy - 4xy = 12$$

$$2xy = 12$$

$$xy = 6$$

Since x and y are distinct integers, the possible pairs (x, y) are (1, 6), (2, 3), (3, 2), and (6, 1).

Calculating the total amount for each pair:

(x, y)	a	b	c	d	Amount (Rs)
(1, 6)	20	10	180	240	450
(2, 3)	40	20	90	120	790
(3, 2)	60	30	60	80	780
(6, 1)	120	60	30	40	750

The only amount that is a multiple of 10 and has a one-digit difference in the number of notes of any two types is 790.

Ravi distributed all his marbles to Mr. X, Mr. Y and Mr. Z such that Mr. X received $k/5$ marbles of the total marbles where k is a positive integer, Mr. Y received 12 marbles and Mr. Z got the remaining marbles. Furthermore it is known that Mr. X has odd number of marbles. If the number of Marbles received by $\boxed{\text{Mr. X} > \text{Mr. Y}}$. Find the sum of all the possible number of marbles Mr. Z can have if Ravi distributed less than 50 marbles.

Handwritten notes and calculations:

$k/5 \rightarrow \frac{1}{5}, \frac{2}{5}, \frac{3}{5}, \frac{4}{5}$
 $k \rightarrow 1, 2, 3, 4$

$5x$
 $\swarrow \quad \downarrow \quad \searrow$
 $x \quad y \quad z$

$3x > 12$
 $x > 4$

$5x < 50$
 $x < 10$

$2x - 12 > 0$
 $2x > 12$
 $x > 6$

x
 $7, 8, 9$

$x = 7$

21	12	2
27	12	6

8 ✓

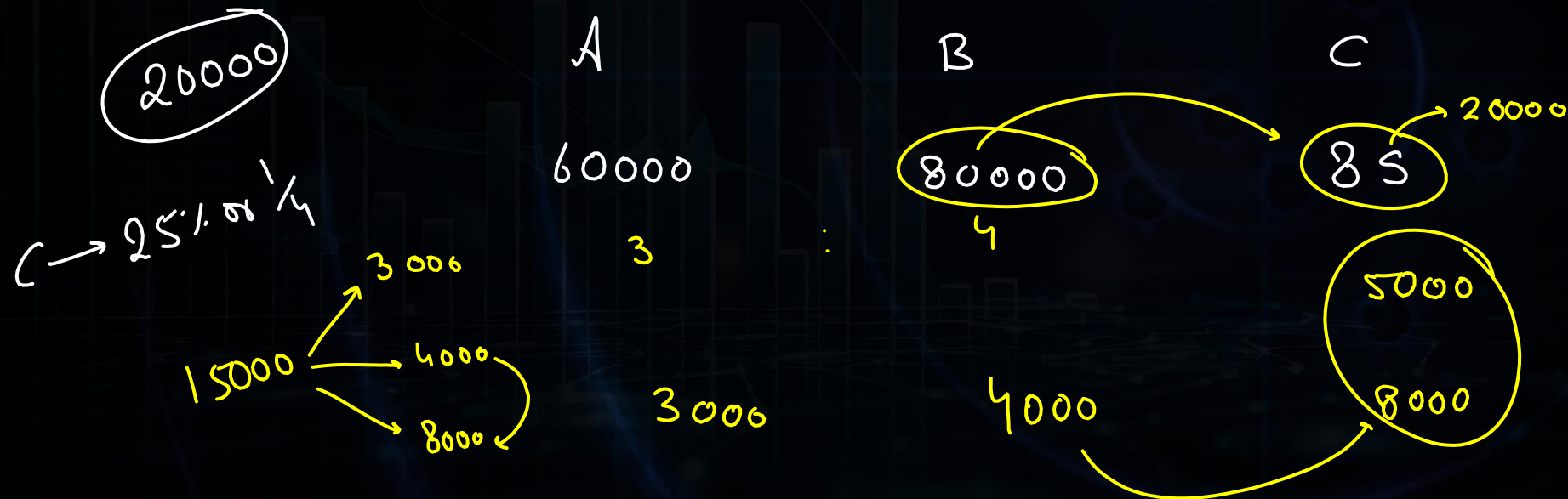
$x > 12$
 $4x - 12$

$2x$
 12

$3x > 12$
 $2x - 12$

$4x$
 12

A, B and C ventured into a new business. A contributed Rs. 10,000 for 6 months, B invested 16,000 for 5 months and C invested a certain sum for 8 months. C manages the entire business for which he takes 25% of the profit and the remaining profit is shared among A, B and C in the respective ratio of their cumulative investments. If at the end of the 8 months the profit realized was Rs. 20,000. Find the sum that C invested, if profit received by B is Rs. 4000.



While testing an electric car on a clear route, it was found that the car runs at a fixed speed when it is unloaded. And the only variable reducing speed was the weight loaded on the car. The reduction in speed was directly proportional to square of the weight loaded on. When the car loaded with 9 kgs, the speed of the car was 60 kmph. And when the was loaded with 12 kgs, the speed of the car was 39 kmph. What would be the maximum integral load that the car can carry in order to reach a warehouse 170 Kms away in at most 10 hour?

Hw

EXTRA SUPPORT



mba wallah

physicswallah.onelink.me/ZAZB/jxe1rd72 and 10 more links

Subscribed

Home Videos Shorts Live Podcasts Playlists Posts

Upcoming live streams

MBA CORE 2026 LRDI CALCULATION TECHNIQUES 01 Upcoming
Calculation Techniques - 01 | LRDI | Free Demo Lec 03 |...
2 waiting
• Scheduled for 10/16/25,
Notify me

LAUNCHING CAT 2025 OFFLINE MARATHON Upcoming
It's Official! CAT 2025 Offline Marathon is LIVE |...
Scheduled for 10/17/25, 12:00PM
Notify me

MBA PREPARATION | CAT 2025

QUANTS MBA POWER BOOSTER 2025 3 videos
QUANTS | MBA POWER BOOSTER 2025
MBA Wallah • Playlist
View full playlist

VARC MBA POWER BOOSTER 2025 6 videos
VARC | MBA POWER BOOSTER 2025
MBA Wallah • Playlist
View full playlist

LRDI MBA POWER BOOSTER 2025 4 videos
LRDI | MBA POWER BOOSTER 2025
MBA Wallah • Playlist
View full playlist

COMPLETE ARITHMETIC AVERAGE 15 videos
Complete Arithmetic for CAT 2025/2026 | Quantitative ...
MBA Wallah • Playlist
View full playlist

Complete Arithmetic for CAT 2...

MBA Wallah - 1 / 15

COMPLETE ARITHMETIC AVERAGE 1:41:36

Algebra | Arithmetic Ep. 15 | CAT 2025 Prep
MBA Wallah

2 COMPLETE ARITHMETIC DISTANCE TIME & SPEED 2:11:26

Distance, Time & Speed | Arithmetic Ep. 14 | CAT 2025...
MBA Wallah

3 COMPLETE ARITHMETIC DISTANCE TIME & SPEED 2:22:39

Distance, Time & Speed | Arithmetic Ep. 13 | CAT 2025...
MBA Wallah

4 COMPLETE ARITHMETIC DISTANCE TIME & SPEED 53:09

Distance, Time & Speed | Arithmetic Ep. 12 | CAT 2025...
MBA Wallah

5 COMPLETE ARITHMETIC S.I & C.I 3:18:51

Simple & Compound Interest (SI CI) One Shot | CAT 2025...
MBA Wallah

6 COMPLETE ARITHMETIC DISCOUNT MRP & FAULTY WEIGHT 58:57

Discount MRP and Faulty Weights | Arithmetic Ep. 10 |...
MBA Wallah

7 COMPLETE ARITHMETIC PROFIT & LOSS 1:03:35

Profit and Loss | Arithmetic Ep.09 | CAT 2025 Prep
MBA Wallah

8 COMPLETE ARITHMETIC REMOVAL & REPLACEMENT 1:09:13

Removal and Replacement | Arithmetic Ep. 08 | CAT 2025...
MBA Wallah

Around
23 Hours Videos

Around
150+ Questions

Concepts
50+ Detail



THANK YOU

